

Curriculum

All the slides and lectures are part of the syllabus. This includes the following topics (in bold the most common topics in the exam):

3D modelling:

- Parts and assemblies
- **Extrusion and revolution**
- **Symmetry and linear and circular patterns**
- Assemblies
- Drawings

3D printing:

- **Advantages and disadvantages**
- **Different 3D printing techniques: FDM, SLS, SLA**
- FDM basic settings: Layer height, infill, speed, support material, etc.

Flat sheet cutting:

- **Advantages and disadvantages**
- **Different flat sheet cutting techniques: Laser cutting, water jet, plasma cutting**
- Basic settings for laser cutting: Speed, power and frequency
- Techniques for flat sheets: finger joints, bending, etc.

Machine elements:

- Structural elements
- **DOF concept**
- **Linkages**
- **Plain and rolling bearings**
- **Linear guides**
- **Power transmission: Gears, pinion and rack, pinion and endless gear, lead screw and belts**

Electronics:

- **Voltage, current and resistance**
- **How to use a breadboard and a multimeter**
- **Ohm's law and voltage dividers**
- **Diodes and LEDs, potentiometers, IR sensors, US sensors and switches: How they work, the circuits needed and how to wire them**
- **DC motors, stepper motors and servo motors: How they work, the circuits needed and how to wire them. Advantages and disadvantages of each of them.**

- **Transistors and H-bridge circuits**
- Voltage regulators and DC/DC converters

Microcontrollers:

- Arduino and microcontrollers
- **Reading sensors with a microcontroller: Circuits and programming**
- **Using actuators with a microcontroller: Circuits and programming**
- Digital communications: SPI and I2C

PCBs:

- **Advantages and disadvantages (compared to a breadboard)**
- Components of a PCB: pads, tracks, vias, etc
- Soldering

Milling and Turning:

- **Advantages and disadvantages**

Moulding and casting:

- **Advantages and disadvantages**
- **Moulding design and its properties**

Study Plan

Exam format

- ~15 min **group presentation** (all members present) + ~15 min **individual Q&A** per student
- Three question themes: **Q1 Manufacturing, Q2 How a machine works, Q3 Electronics**
- Q3 includes **practical wiring** on a breadboard — practice this; many students fail basic circuits
- Bring your **prototype** and a **smartphone** as a backup display

Study **bold curriculum topics** first, then cover the rest for breadth.

Study progression

Work through these in order. Each step builds on the last; don't skip to electronics before you can explain your own prototype's mechanics.

1. Learn the theory (slides + bold topics)

- ☒ 3D modelling: extrusion, revolution, symmetry, patterns
- ☒ Manufacturing methods: 3D printing (FDM/SLS/SLA), flat-sheet cutting (laser/water jet/plasma) — pros, cons, and when to use each
- ☐ Milling/turning, moulding — pros, cons, and when to use each
- ☒ Machine elements: DOF, linkages, bearings, linear guides, power transmission (gears, belts, lead screw, rack & pinion)
- ☒ Electronics fundamentals: voltage/current/resistance, Ohm's law, voltage dividers, breadboard, multimeter
- ☒ Components: switches, LEDs, potentiometers, IR/US sensors, diodes, DC/stepper/servo motors, transistors, H-bridge, ultrasonic sensors, voltage converters
- ☒ Microcontrollers: reading sensors, driving actuators, basic SPI/I2C
- ☐ PCBs: pros/cons vs breadboard, pads/tracks/vias, soldering basics

2. Practice Q1 — Manufacturing

For parts from your project, lecture slides, or exam examples, answer all three variants:

- ☐ Default: what process and material, and why?
- ☐ “What if it must be metal?”
- ☐ “What if you need 1000 units?”

Always justify with cost, time, tolerance, material properties, and surface finish.

3. Practice Q2 — How a machine works

Apply this sequence to your prototype and a few machines from the slides:

- ☐ Identify components: linear guides, bearings, motors, belts/gears/screws, structural elements
- ☐ Trace power: motor -> reduction (if any) -> end effector
- ☐ Count DOFs
- ☐ Name sensors and actuators — which type, and **why**?
- ☐ Identify driver (H-bridge, stepper driver) and microcontroller

4. Practice Q3 — Electronics

- ☐ Explain how to read a switch and a potentiometer (circuit + code)
- ☐ Explain how to control a DC motor (direction and speed) and how a servo works internally
- ☐ Wire from memory on breadboard or simulator: button, LED with resistor, potentiometer, DC motor via H-bridge, servo
- ☐ Draw schematics with correct symbols for every component used in the course
- ☐ Redo relevant MAs until wiring is automatic

5. Tie it together on your prototype

- ☐ For each part: manufacturing choice (Q1)
- ☐ Full mechanism walkthrough (Q2)
- ☐ Complete electronics schematic and wiring (Q3)
- ☐ Rehearse the group presentation; each member can answer “why did you choose X?”

6. Final check before the exam

- ☐ Prototype working (or ready to explain failures clearly)
- ☐ Can wire a button circuit on a breadboard without help
- ☐ Can justify every design and manufacturing decision on your project
- ☐ Smartphone charged; presentation parts assigned

Notes on Theory

3D modelling: extrusion, revolution, symmetry, patterns

Term	Description
Extrusion	Create a 3D solid by pushing or pulling a 2D sketch profile along a straight path (distance and direction). Used for prismatic parts like brackets, boxes, and beams.
Revolution (revolve)	Create a 3D solid by rotating a 2D profile around an axis. Used for axisymmetric parts like shafts, wheels, and bottles.
Chamfer / Round	Select one or more edges and round them to a certain circular radius / chamfer with an angle and distance
Symmetry (mirror)	Duplicate geometry across a mirror plane so you only model half the part. Keeps the model parametric and consistent — change one side and the other updates.
Linear pattern	Copy a feature or part at equal spacing along a straight line (e.g. a row of holes or repeated ribs). Also Rectangular pattern in 2 dimensions with the same idea.
Circular pattern	Copy a feature or part at equal angular spacing around an axis (e.g. bolt holes on a flange or teeth on a gear blank).
Part	A part is a single fusion file representing one object.
Assembly	An assembly is a collection of parts fit together to resemble a bigger gathered object.

Term	Description
(Technical) Drawing	A technical drawing is a 2D representation of a 3D part and/or assembly. It can be used as the basis for laser cutting or in order to mass produce a given 3 dimensional part by supplying all relevant dimensions.
CAD	Computer Aided Design - refers to using software to design parts, such as we do in Fusion360 when making parts and assemblies
CAM	Computer Aided Manufacturing - refers to using software to manufacture parts, such as we do in Prusa Slicer to translate our 3D designs into a programme for the Prusa 3D printer to follow in order to manufacture our part. It also encompasses things such as technical drawings.
Fully constrained sketch	Every point and line in the sketch has its position and size fully defined (dimensions, angles, coincident/tangent/parallel constraints). Unconstrained geometry is blue; fully constrained turns white (in dark mode). Prevents accidental shape changes when editing.
Parameterised model	Key dimensions are defined as named parameters (variables) rather than hard-coded numbers. Change one parameter (e.g. wall thickness) and the whole model updates automatically — essential for iteration and design variants.

Product Design

Design a product based in categories, describing the product from each of the following perspectives:

- *Interface*: Type and location of sensors and any user interface (UI)
- *Mechanism*: What kinds of mechanisms make the product function / behave in the desired way?
- *Outputs*: Which motors are used (strongly coupled to mechanism) and other elements (screens, lights, etc.)
- *Power*: How is the product powered? (Wall Plug, Batteries, Solar Cells etc.)
- *Size*: What is the physical size of the product? (height, breadth, depth, weight?) Can also be defined in terms of a specific product-relevant metric such as # of products.
- **Requirements**: What are the functional requirements for the behavior of the product?

A traditional workflow in a CAD programme contains: - Select a 2D plane in

3D space to make a sketch on - Draw a sketch on the plane - Extrude / Revolve
/ Cut the sketched shape into a 3D part - Repeat

Extra Takeaways

- The earlier in the process we can make a working model, the better - the further we get in the production chain, the more expensive it is to iterate and change the product.
- Make sure your sketches are **fully constrained**. Unconstrained sketches drift when you edit later features, and fully constraining makes edits easier to follow.
- If creating a hole for a screw / shaft, add some clearance to the diameter (of ~nozzle width) to ensure the part will fit through the manufactured part! (f.ex. 3.3 mm is good for M3 screws)

Technical Drawings

A technical drawing is a 2D representation of a 3D part. It is used for multiple purposes, hereunder illustration, description, documentation and the basis for manufacturing parts in a workshop. We can represent a part in multiple ways:

- *Projection*: refers to a perspective on the 3D object in 2D space. It may represent a specific angle or a top view.
- *Multi-view Projection*: Aims at providing a full description of an object by showing the flat surfaces from multiple perspectives (top, bottom, front, back, right, left) - often aimed at describing the object fully with the least amount of sketches.
- *Exploded-view Drawing*: Displays multiple parts, that are assembled together, exploded apart such that one can examine each individual part of the assembly.

Rules for adding dimensions on the drawing: - Add **all** dimensions - **Display** dimensions in the most descriptive view of the feature. - Dimension lines shall **never cross other lines**

Manufacturing methods: 3D printing (FDM/SLS/SLA), flat-sheet cutting (laser/water jet/plasma), milling/turning, moulding — pros, cons, and when to use each

Here is a high level overview of the pros and cons of the main types of manufacturing. Detailed pros / cons of subtypes are described in the following sections.

Method	Description	Pros	Cons
3D printing	Builds parts layer-by-layer from a digital model (additive — material is added, not cut away). FDM, SLS, and SLA are the main subtypes.	Only little waste, can create complex parts (infeasible assemblies / hollow parts), good for prototyping because of short lead time	Layer lines and weaker strength between layers - less resistant than raw material, slow and costly at high volume
Flat-sheet cutting	Cuts 2D profiles from flat stock (acrylic, plywood, metal plate, etc.) along a programmed path. Main methods: laser (melts/vaporises), water jet (high-pressure abrasive water), and plasma (ionised arc through metal).	Fast for flat parts; low setup cost and minimal fixturing; fine detail; good for enclosures, panels and finger-joint boxes; can combine with bending for 3D shapes	2D profiles only — true 3D needs bending, stacking, or assembly; thickness and material limits depend on method.
Milling	A rotating cutter removes material from a solid block (subtractive CNC).	High precision and tight tolerances; wide material choice including metals; excellent surface finish and strength	Material waste; slow and expensive for complex 3D shapes; skilled setup required; poor choice for large production runs of simple parts

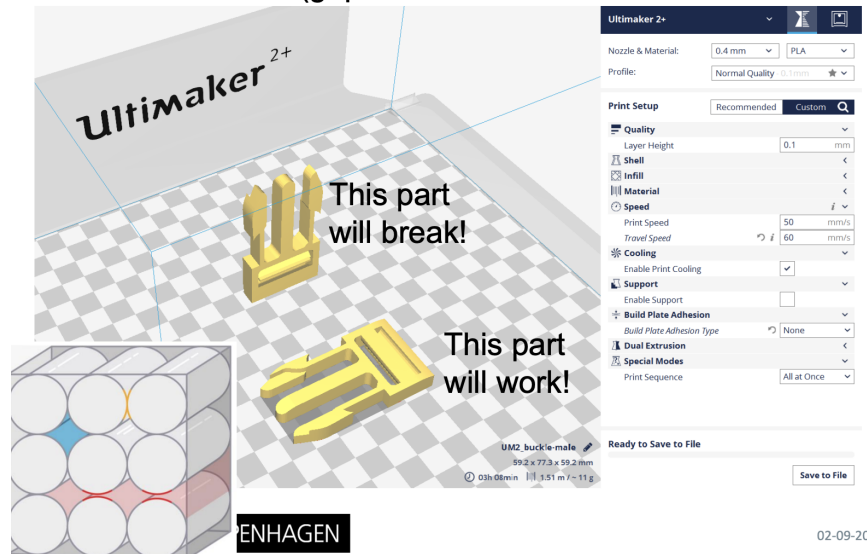
Method	Description	Pros	Cons
Turning	The workpiece rotates on a lathe while a fixed cutting tool shapes it — for axisymmetric parts.	Very efficient for round parts (shafts, pins, bushings); high precision on cylindrical features; good surface finish	Only axisymmetric (round) parts; material waste; setup time; usually overkill for quick plastic prototypes
Moulding & casting	Liquid material is poured or injected into a mould cavity and solidifies (e.g. injection moulding, silicone casting).	Lowest unit cost at high volume (1000+ parts); consistent, repeatable parts; complex shapes in one step	High upfront mould/tooling cost; long lead time to make the mould; design changes are expensive; mainly suited to specific materials (plastics, metals, silicone)

3D printing types w. pros & cons

Traditional manufacturing was *subtractive*, meaning you start with a larger block of material and cut away the parts that are not needed in order to obtain the desired shape. 3D printing is, on the other hand, an *additive* manufacturing method, meaning we add material until reaching the desired shape.

Method	Description	Pros	Cons
LOM (<i>Laminated Object Manufacturing</i>) / SDL (<i>Selective Deposition Lamination</i>)	sheets are successively glued together on top of each other and cut with a knife / laser	Cheap materials, good for mock-ups	layered surfaces, weakness between layers, waste from cutting layers
SLS (<i>Selective Laser Sintering</i>)	A thin layer of powder is distributed on the printing surface and a laser is used to <i>sinter</i> the sections that should be melted together into the manufactured part. A new layer is then distributed on top and process is repeated.	loose powder acts as support structure	Grainy surfaces, post processing can be difficult / expensive
SLA (<i>Stereolithography</i>)	A laser is used to cure resin in a layer-by-layer fashion on a surface that moves upwards.	More sturdy and elastic results, different material options, high quality and resolution	Expensive, need support material
SLS (<i>Fused Deposition Modelled</i>)	Melting and placing material (PLA, ABS, TPE) in layers on a build surface, building the part line-by-line	Cheap and accessible, good for prototyping	Anisotropy (stronger on one axis than on other ones due to layers), Needs support, layered surfaces

XY directions are strong (long filaments),
Z direction is weak (gaps and weak adhesion between filaments)



Anisotropy Explained

Slicers and Settings Slicers creates a stack of 2D representations of the 3D model, which results in the 3D model! This sections provides an overview of the settings that are relevant to remember:

- *Layer Height* (Default: nozzle diameter / 2) defines the distance in the z-direction between each slice in the stack. Large values will give low resolution, but default is often good enough for most prototyping work!
- *Infill* (Default: 5% or 10%) defines how much filament should be filled on the inside of parts. Adds stability, but is a waste of material if it is not needed.
- *Speed* (Default: 15-30 mm/s) defines how quickly the printer head will move. Higher speeds risk dragging the material so it is not properly extruded where it is supposed to be.

Common Printing Problems

- *Adhesion* (The base layer is not sticking and has gaps or is tumbled over) - the bed is not leveled, the bed is oily or otherwise dirty - use brim / raft (area around base is printed to support the main print)
- *Warping* (The base layer is misshaped) - bed is not heated, fan is cooling of too quickly - use brim / raft (area around base is printed to support the main print)

Flat Sheet types w. pros & cons

Flat-sheet cutting is a *subtractive* process: you start with a flat sheet and cut away material along a 2D path. Unlike 3D printing, you cannot build overhangs — but you can bend, stack, or join cut pieces (finger joints) to make 3D structures.

Method	Cuts	Speed	Quality	Max .thickness	Other
Plasma cutting	Only metals	Fast	Low	< 20mm	Fumes
Water jet	Everything, not engraving	Slow	Good	< 250mm	No heat
Laser cutting	some materials (depends on the laser)	Medium	Good	< 12mm	Fumes

Method	Description	Pros	Cons
Laser cutting	A focused laser beam melts or vaporises material along a programmed 2D path.	Fast on thin sheets; good for acrylic, plywood, enclosures, and finger-joint boxes; minimal fixturing	Thickness and material limits; kerf width loses material; fumes from some plastics (need ventilation); reflective/thick metals need a more powerful laser
Water jet cutting	High-pressure water, often mixed with abrasive, erodes material without heat.	No heat-affected zone — no warping; cuts thick and hard materials (metal, stone, glass); high precision; no toxic fumes	Slower than laser on thin sheets; messy; wider kerf; higher machine running cost

Method	Description	Pros	Cons
Plasma cutting	An ionised gas arc melts through conductive sheet metal, blown away by compressed air.	Fast and relatively cheap for thick steel and other conductive metals; good for heavy plate	Rougher edge finish than laser; heat-affected zone; conductive metals only;

Laser settings The main settings to remember for laser cutting:

- *Speed* — how fast the head moves. Higher speed = faster cut but may not cut through; too slow can over-burn the material.
- *Power* — laser intensity. Higher power needed for thicker or denser materials.
- *Frequency* — pulses per second (on some machines). Affects cut quality on certain materials, especially metals.

Flat sheet techniques

- *Finger joints* — interlocking rectangular teeth cut into sheet edges so two pieces slot together without fasteners. Common in laser-cut plywood boxes.
- *Kerf* — the width of material removed by the cut. Account for kerf in your CAD so assembled parts fit (typically ~0.1–0.2 mm for laser).
- *Bending* — shaping a flat 2D sheet into a 3D shape
 - Metals: Bending deformation (a manual machine applying forces)
 - Plastic: Heat bending (on some types)
 - Paper / Cardboard: Folding, possibly with cut hinges

Flat Sheet Materials

- *Plywood* - expensive, thicker (should ask lab staff before using it)
- *MDF* - Medium Density Fiberboard (this is what we used!) - brittle & expands with water
- *Cardboard* - weak and quickly becomes useless with water / screws etc.
- *Acrylic* - easy to laser cut but brittle!
- *POM* - difficult to laser cut, expensive, low friction

Machine elements: DOF, linkages, bearings, linear guides, power transmission (gears, belts, lead screw, rack & pinion)

Objects move in space based on **forces** applied. **Restrictions** may change how the forces are being actuated in the environment.

A mechanism is formally defined as: *a combination of rigid or resistant bodies, formed and connected [with kinematic pairs] so that they move with definite relative motions with respect to one another* - i.e. it exists with or without motors and simply define bodies that move relative to others bodies.

A machine extends on this with motors, by defining it to also include *transmit force from the source of power to the resistance to be overcome*

Degrees of Freedom

One of the most important concepts to understand; usually **DOF = # Motors**, since each possible movement direction in a mechanical part should usually be controlled by driving force. Formally: *Number of independent parameters that define the configuration of a machine (position and orientation of all the components)* - some machines have many (plane) while others have few (train on train tracks).

Mechanisms (DOFs)



In this case bodies are usually represented as a line or triangle and revolute joints as a circle.

$$M = \text{degrees of freedom} = 3(n - 1) - 2j_p - j_h$$

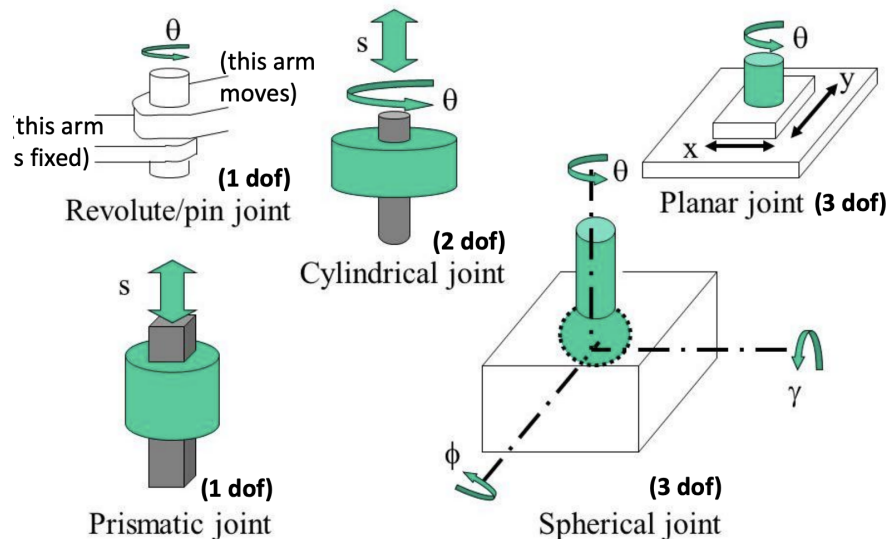
n : n° of links in the mechanism (including the fixed link)
 j_p : n° of 2-dof joints (cylindrical, fork, etc.)
 j_h : n° of 1-dof joints (revolute, etc.)

Joints & Linkages

Define a pair of two bodies, that constrain the movement of the other. See specific examples and their DOFs in below image:

Joints - Kinematic pairs

Connection between two bodies that imposes constraints on their relative movement



Bearings Formally: *A machine element that constrains relative motion to only the desired motion, and reduces friction between moving parts* - i.e. ensures with a fixed object that another body follows a desired motion pattern in space relative to the fixed object.

Two main types: **Plain Bearings** and **Ball Bearings** (Rolling bearings). Ball bearings produce less friction but require lubrication and are more expensive than plain bearings.

Guides An element that constrains relative motion to 1 DOF. Particularly, linear guides are used in many applications, such as 3D printers, to limit motion to a single linear direction. Types of linear guides:

- **Rail + Carriage:** either plain or with rolling bearings, a rail defines the linear direction and the carriage moves along the rail which it surrounds.
- **V-slot:** a part is fastened into the v-slot from which it can only move in the linear direction along the slot.

Structural Elements

Are the parts of our mechanism that do not move and should remain still. Remember that joints are weak and triangular patterns make them slightly

Types of screws

Wood Screws
Machine Screws
Socket Screws
Set Screws

Avoid wood screws
Use the right screwdriver
Check the screws that you need
Are in the lab
Fit in your design

Diagram illustrating various types of bolt heads and their corresponding labels:

- FULL BEARING
- WASHER FACED
- DOUBLE CHAMFERED
- SQUARE
- SLOTTED HEXAGON
- COUNTERSUNK
- RATED COUNTERSUNK
- ROUND
- PAN
- CHEESE
- RATED-CHEESE HEAD
- HEXAGON (TUBULAR)
- 'H' TORQUE
- PHILLIPS COUNTERSUNK
- PHILLIPS RATED COUNTERSUNK
- PHILLIPS ROUNDPAN
- COUNTERSUNK HEXAGON SOCKET
- BUTTON
- FLAT OR CAP HEXAGON
- ROUNDED HEAD

Wood Screws
Screws with a smooth shank and tapered point for use in wood. Abbreviated WS

Machine Screws
Screws with threads for use with a nut or tapped hole. Abbreviated MS

Thread Cutting Machine Screws
Machine screws with a thread cutting (self tapping) point.

Sheet Metal Screws
Fully threaded screws with a point for use in sheet metal. Abbreviated SMS

Self Drilling SMS
A sheet metal screw with a self drilling point.

Hex Bolts
Bolts with a hexagonal head with threads for use with a nut or tapped hole. Abbreviated HHMB or HBXT.

Carriage Bolts
Bolts with a smooth rounded head that has a small square section underneath.

Lag Bolts
Bolts with a wood thread and pointed tip. Abbreviated Lag.

Socket Screws
Socket screws, also known as Allen Head, are fastened with a hex Allen wrench.

Set Screws
Machine screws with no head for screwing all the way into threaded holes.

Eye Bolts
A bolt with a circular ring on the head end. Used for attaching a rope or chain.

Eye Lags
Similar to an eye bolt but with wood threads instead of machine thread.

J-Bolts
J shaped bolts are used for tie-downs or as an open eye bolt.

U-Bolts
Bolts in U shape for attaching to pipe or other round surfaces. Also available with a square bend.

Shoulder Bolts
Shoulder bolts (also known as stripper bolts) are used to create a pivot point.

Elevator Bolts
Elevator bolts are often used in conveyor systems. They have a large, flat head.

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Thread Rods may also be utilized as a cheap lead screw for a rail + carriage implementation or to otherwise be used as a structural element.

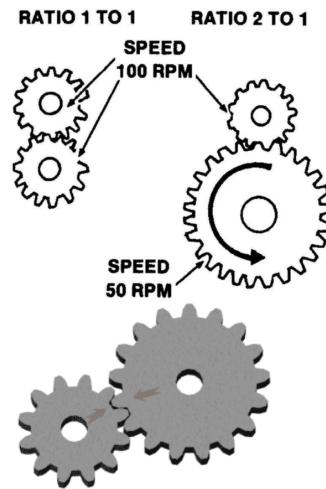
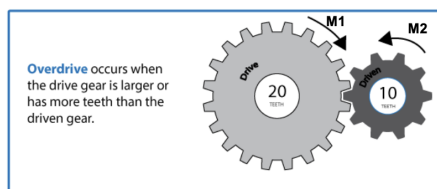
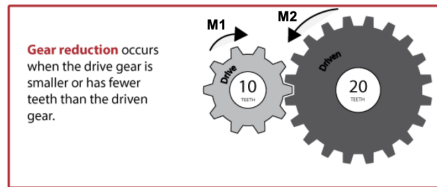
A *force* is a measure of how much relative acceleration is applied to an object by another object. *Gravity* and *Friction* are two common forces that we always encounter everywhere. Formula: $F = m * a$ (Force = Mass * Acceleration) - as a vector in 3D space!

Here we list the main types of power transmission and their features:

Method	Motion	Description	Notes
Gears	Rotational <-> rotational	Two meshing toothed wheels transfer rotation between shafts. Gear ratio = driven teeth / driving teeth — sets speed and torque trade-off.	Compact and efficient; precise ratio; can change direction. Noisy at high load; requires precise alignment; distance between shafts is fixed by gear size
Pinion and rack	Rotational <-> linear	A small rotating gear (pinion) meshes with a straight toothed bar (rack). Pinion rotation drives the rack linearly — or linear rack motion spins the pinion.	Used in CNC axes, steering systems, and moving a bed along a rail; rack must be straight and supported
Pinion and endless gear	Rotational -> rotational	The rack is bent into a circle (ring / endless gear); a pinion drives around its inner or outer teeth. Pinion spin rotates a platform or turntable continuously.	Used for spinning trays, indexing tables, and turntables. Smooth continuous rotation; limited load capacity; self-locking and not very efficient;

Method	Motion	Description	Notes
Lead screw	Rotational $\leftrightarrow$ linear	A threaded screw turns inside a nut (or nut travels along a fixed screw). Each rotation moves the nut a fixed linear distance (pitch of the thread).	High mechanical advantage; precise positioning; self-locking (won't back-drive under load).; Used in 3D printer Z-axes, vices, and linear actuators; Slower than belts; friction and wear on threads
Belts	Rotational $\leftrightarrow$ rotational, linear $\leftrightarrow$ linear, or rotational $\leftrightarrow$ linear	A flexible belt runs over pulleys to transfer motion. Rot $\leftrightarrow$ rot: belt between two pulleys (like a bike). Rot $\leftrightarrow$ linear: one pulley drives a belt that pulls a carriage in a straight line (common on 3D printer gantries). Linear $\leftrightarrow$ linear: belt loop moves a point along a fixed path.	Quiet; can span long distances; cheap; absorbs shock; less precise than gears or lead screws

Gears



$$\text{Gear ratio} = \frac{\# \text{Teeth output gear}}{\# \text{Teeth input gear}}$$

$$M2 = \text{Gear ratio} * M1$$

$$\text{Speed2} = \frac{1}{\text{Gear ratio}} * \text{Speed1}$$

Types of Gears



Spur Gear



Helical Gear



Double Helical Gear
(Herringbone Gear)



Spiral Bevel Gear



Miter Gear



Straight Bevel Gear



Internal Gear



Worm Gear



Rack and Pinion

Electronics fundamentals: voltage/current/resistance, Ohm's law, voltage dividers, breadboard, multimeter

Term	Definition	Description	Notes / Comments
Voltage	V	Electrical <i>Potential</i> Energy - It is a RELATIVE measure, often measured relative to GND (0V)	Potential is the key here; consider it as the height of water in a water tank
Current	A / I	Size of <i>Flow</i> of Electric Charge	This is how fast the flow of electric charge is. Higher current = more charge moving through at each point in time
Resistance	R / Ω	Amount of Difficulty to <i>pass</i> the <i>current</i> through a conductor	I.e. how much “push power” is needed for the current to move through this obstacle, can also analogously describe the width of the connector between two water tanks of different V
Ohm's law	$V = R * I$	Describes the relationship between voltage, current and resistance	
Kirchhoff's First Law	$\sum I_i = 0$ in junction	Describes how everything that flows INTO a junction also flows OUT OF a junction	
Kirchhoff's Second Law	$\sum V_j = 0$ in closed loop	Describes how the sum of voltages around a circuit correspond to the total Voltage from GND to V++	
Parallel Resistors	$\frac{1}{R_{tot}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$		

Term	Definition	Description	Notes / Comments
Serial Resistors	$R_{\text{tot}} = R_1 + R_2 + \dots$		

Physical Item	Description	Notes / Comments
Power Supply	Used to supply power to a circuit (kV & GND)	Will usually use CV (Constant Voltage) and a maximum Current (rather than the opposite)
Multimeter	Used to measure things in circuits - both connectedness, voltage, current, resistance and more	To measure <i>Current</i> , connect the multimeter <i>In-Series</i> . To measure <i>Voltage</i> , connect the multimeter <i>In Parallel</i> for the two points of the Voltage drop measurement.
Breadboard	Outer Power rails are connected horizontally, inner rails are connected vertically	Used to build circuits for prototyping and experiments

Circuit Name	Description	Notes / Comments
Ohms Law Circuit	Connect a Power supply to a breadboard with a resistor. Measure current & voltage using a multimeter (parallel and in-series as applicable). Disconnect the power supply and measure resistance with the multimeter.	Used to convince yourself that Ohm's Law is true.
Voltage divider	With two resistors in parallel we can tune the voltage going out in-between the resistors	$V_{out} = V_{in} * \frac{R_2}{R_1 + R_2}$ (how much relative voltage is left for the second resistor out of the total resistance in the circuit)

Representing Circuits

A circuit can be represented in many ways. The most common ways are the following:

- **Schematic:** symbolic and cleaner representation of a circuit. Lines do not represent wires, but represent a connectedness between voltage levels.
- **Breadboard / Wiring:** physical representation of the circuit - how it is actually wired in order to work - these are often messier and difficult to read.
- **Printed Circuit Board:** physical implementation of circuit in a production or advanced prototyping.

Components: switches, LEDs, potentiometers, IR/US sensors, diodes, DC/stepper/servo motors, transistors, H-bridge

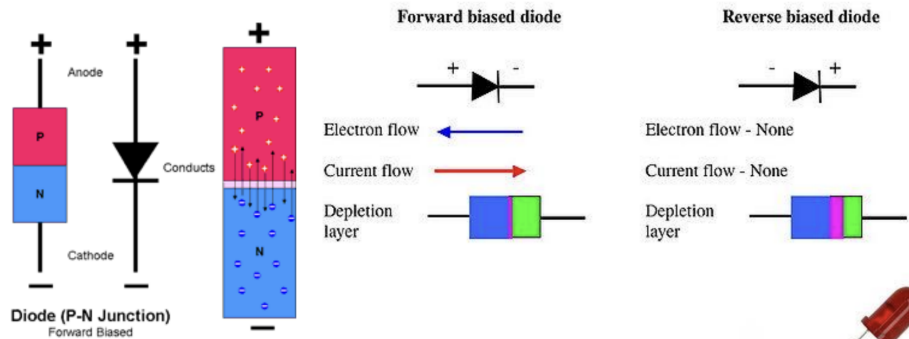
Component	Description	Example Usage	Pins
Potentiometer	Functions as a voltage divider, tuning the V_{out} on the signal pin. Reading a potentiometer is done by measuring the voltage between the signal and GND.	changing the intensity of a light bulb	V++, GND, Signal
Diodes	A polarized component, if forward-biased, current will flow, and the Voltage will be static at ~2.1V	Connected in a circuit along with resistors (to sustain the remaining Voltages)	V++, GND
LEDs	A specific type of Diode, emitting light when current flows	Light an LED in a circuit with power, LED and a smaller resistor	V++, GND
Switches / Buttons	Connects or disconnects two points in a circuit. Can have 2 or more terminals. Can also be a switch between two states.	Connect a switch to the LED circuit to only have current flowing to the LED when it is pressed.	V++, GND (but needs high resistance on either side!)

Component	Description	Example Usage	Pins
Optosensor	An optosensor describe any sensor that uses infrared light for detection. Two main categories: <i>Reflective</i> (sensor and emitter point in the same direction and detect reflection of emitter) and <i>Break beam</i> (sensor and emitter point toward each other and detect when something is prohibiting the signal between them)	IR Sensor or Shaft Encoder	V++, GND, Signal
Shaft Encoder	Shaft encoder are <i>break beam optosensors</i> , which use a plate with holes to switch between detection states.	Can be used to “count” ticks or determine position.	-
IR Sensor	IR sensors are <i>reflective optosensors</i> . It is an <i>active</i> sensor that needs to be powered to work. Digital Out (DO) will send signal if above threshold configured with potentiometer.	Detecting a line	V++, GND, Signal
Transistor	Used to switch on / off based on a Voltage signal for handling big currents! (i.e. amplifying a signal current)	Amplifying IR Sensor Signal to drive Motor. Implementing logic gates.	Collector (5V load), Emitter (GND) & Base (Signal)

Component	Description	Example Usage	Pins
H-Bridge	Used to control the direction of current through a motor.	May be used to change direction of DC Motor.	-
DC Motor	Consumes constant voltage and may change direction with the direction of the current. Speed is proportional to Voltage and depends on the LOAD. Torque is proportional to Current ($I > 200\text{mA}$)	Anything that need to move	V++, GND
Darlington Configuration	Transistors are used to amplify the current of a signal such that a motor will run. The darlington configuration simply amplifies it twice using 2 NPN transistors	Amplifying IR Sensor Signal to drive Motor	-

Diode Drawing

A new component: Diode



2 terminal

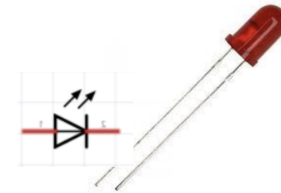
Polarity

Direction 1: the current flows

Direction 2: open circuit (no current)

If current flows $\Rightarrow V \approx 2.1v$

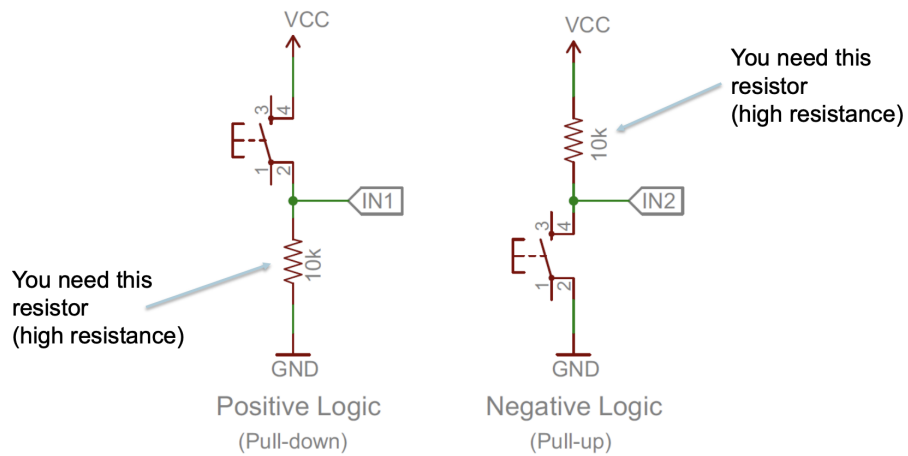
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A LED is a diode that emits light if current flows!

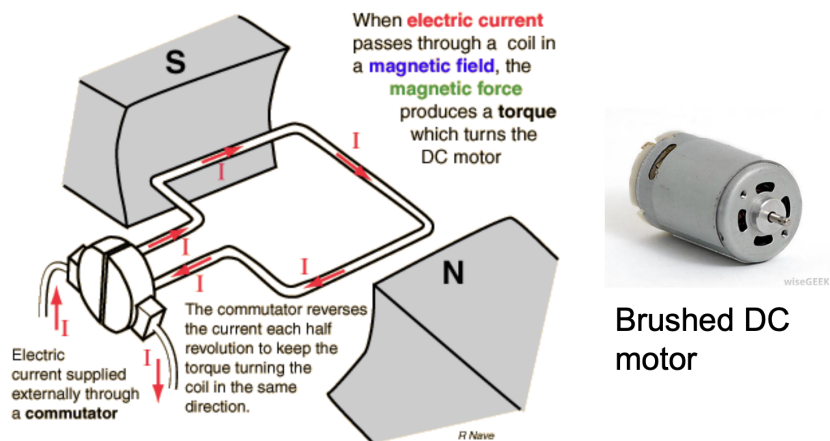
Reading a Switch

How to read a switch?



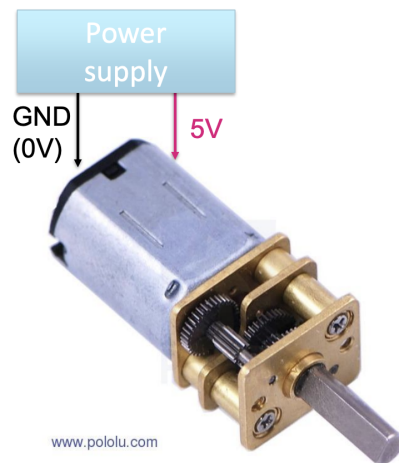
Without the resistor, the voltage is not defined when the switch is not pressed!

DC Motors

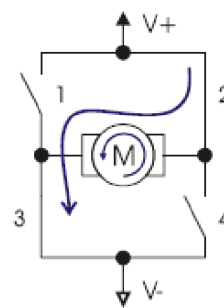
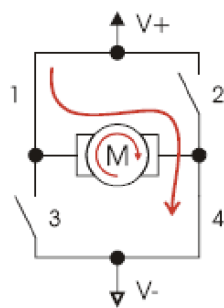
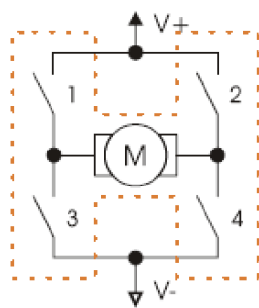


Brushed DC motor

Many variation of this design used

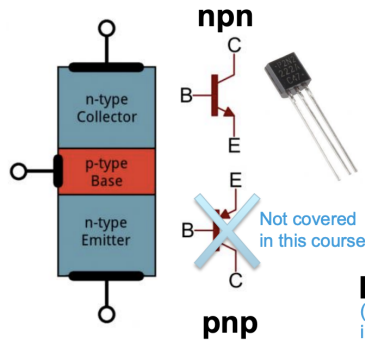


Motor Type	Stall Current @ 6 V	No-Load Speed @ 6 V	Approximate Stall Torque @ 6 V	Single-Shaft (Gearbox Only)
high-power, carbon brushes (HPCB)	1600 mA	3000 RPM	4 oz-in	10:1 HPCB
		1000 RPM	9 oz-in	30:1 HPCB
		625 RPM	15 oz-in	50:1 HPCB
		400 RPM	22 oz-in	75:1 HPCB
		320 RPM	30 oz-in	100:1 HPCB
		200 RPM	40 oz-in	150:1 HPCB
		140 RPM	50 oz-in	210:1 HPCB
		120 RPM	60 oz-in	250:1 HPCB
		100 RPM	70 oz-in	298:1 HPCB
		32 RPM	125 oz-in	1000:1 HPCB



Transistors

Transistors

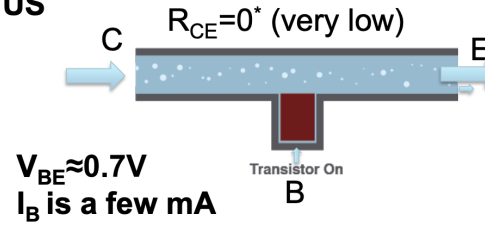


Some of them are able to handle high currents from collector to emitter!

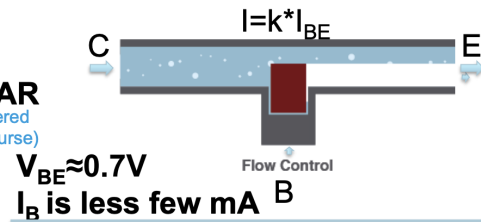
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3 STATUS

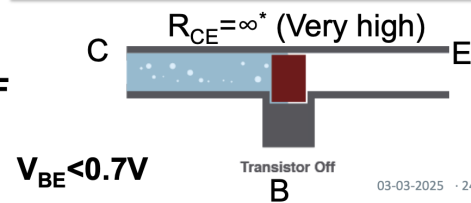
ON



LINEAR
(Not covered in this course)

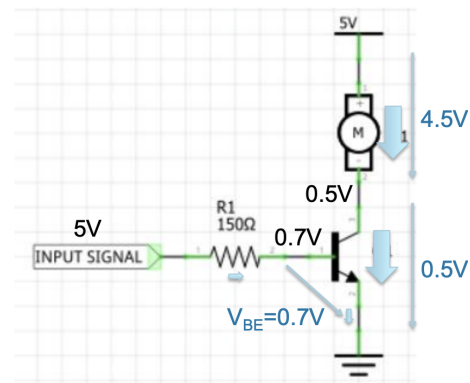
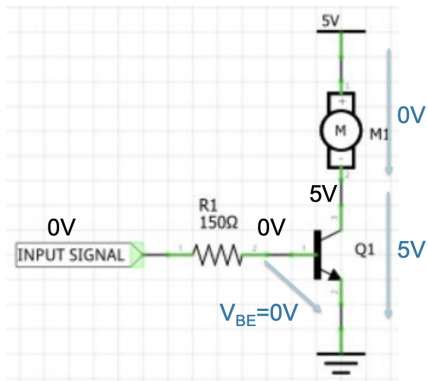


OFF



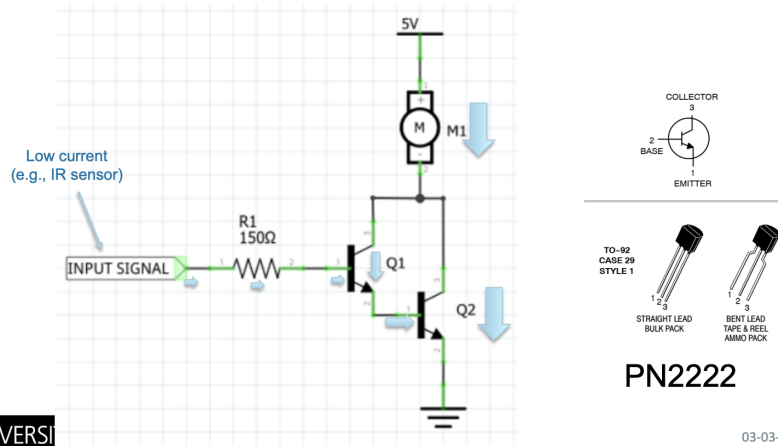
Input signal = 0V

Input signal = 5V



Darlington configuration

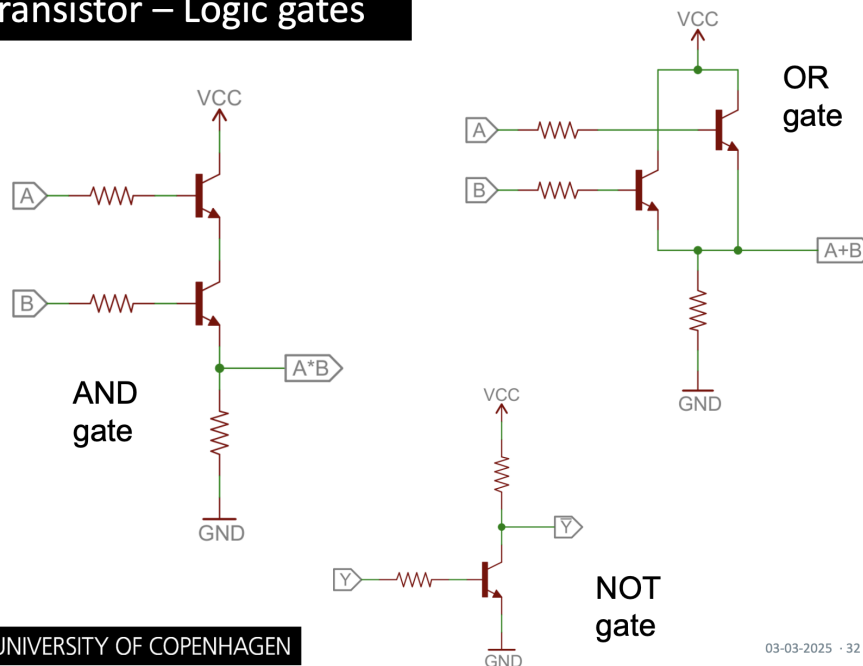
Two transistors: One transistor amplifies the current that is introduced into the base of the second transistor. The second transistor amplifies even more the current.



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Transistor – Logic gates



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Advanced Electronics: Stable Voltage, advanced sensors and motors

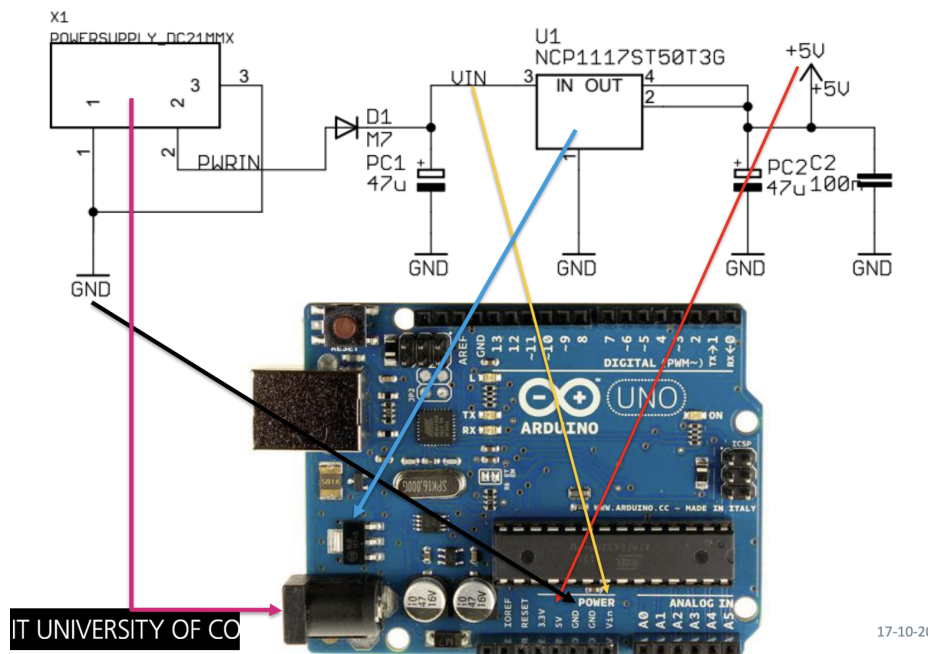
Component	Description	Solves Problem. . .	Cons	Notes
Linear Regulators	use a fixed Voltage reference and a transistor to produce constant output voltage	Input Voltage Variations	Generates Heat when $V_{in} \gg V_{out}$, can only decrease i.e. it must hold that $V_{in} > V_{out}$	Place capacitors for both V_{in} and V_{out} for stability. When powering the arduino with the 12V plug, a linear regulator is used to produce the 5V port.
DC/DC Converter	Converts one DC voltage to another DC voltage. <i>Buck</i> steps down, <i>Boost</i> steps up, <i>Buck-Boost</i> may do either. Almost no energy loss.	Input Voltage Variations	Noisy (not suitable for precise analog measurements)	Power In = Power Out, but the Current (I) changes inversely proportional to the voltage.

Component	Description	Solves Problem. . .	Cons	Notes
Ultrasonic Sensor	Similar to IR sensors but works with sound and time rather than light reflections. Sends a signal through the trigger and measures time passed before receiving the echo.			
Stepper Motor	uses magnets and coils to rotate at known step intervals	move things known relative distances from starting position	needs homing at each restart. Consumes a lot of current at all times.	
Servo Motor	rotates to a desired angular position in its range given a signal	move things between angles with precision and with “memory” (i.e. does not need reset position after restart)	limited motion space	uses a potentiometer to determine its position

Schematics

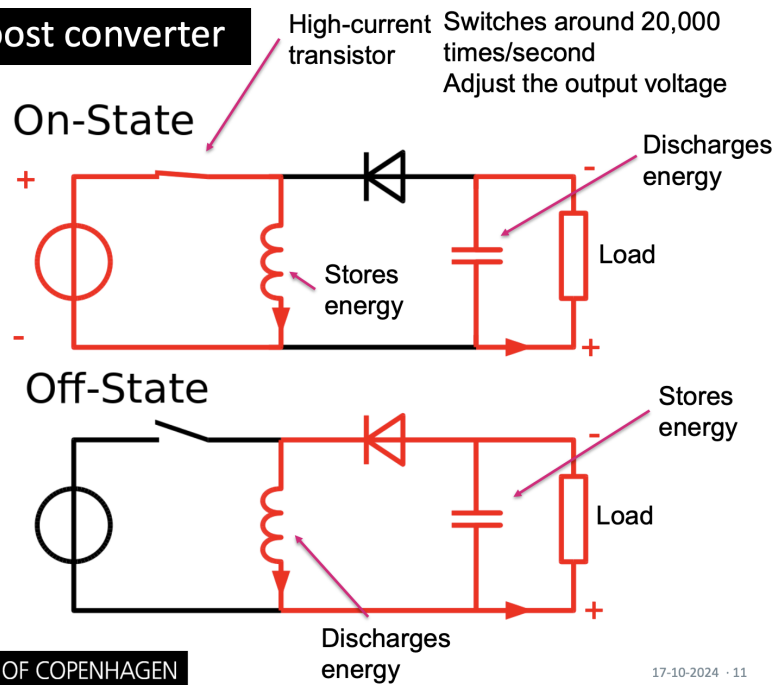
- **Linear Regulator**

Example II: Arduino Uno schematics



- Buck-Boost

Buck-Boost converter

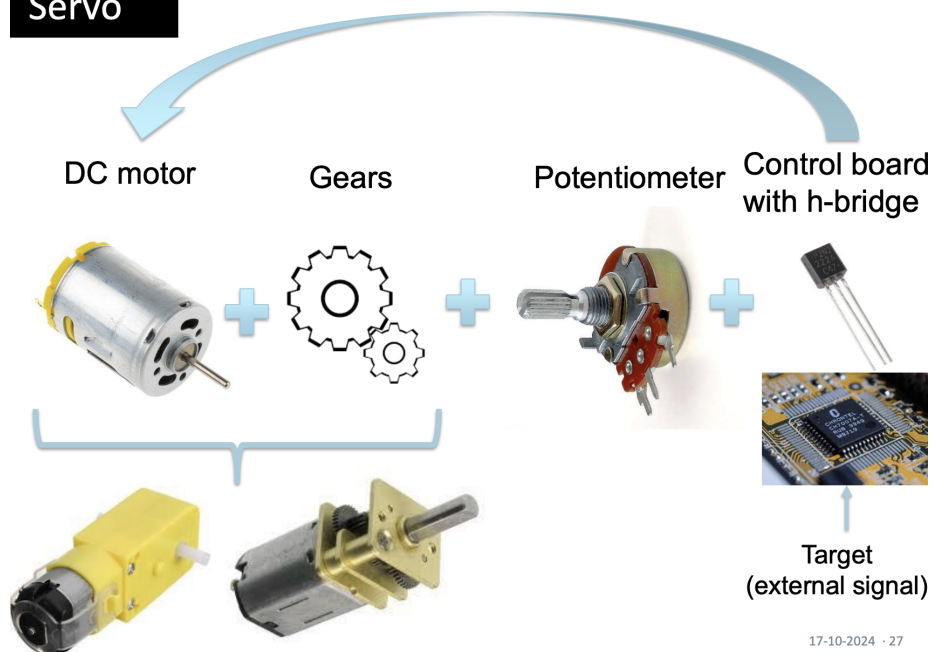


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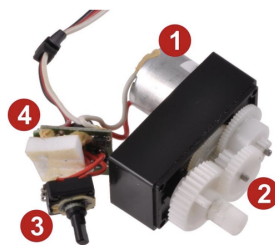
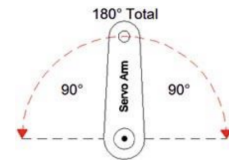
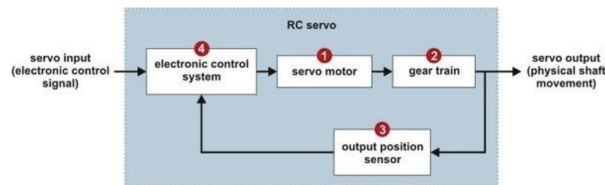
Servo Motor

Servo

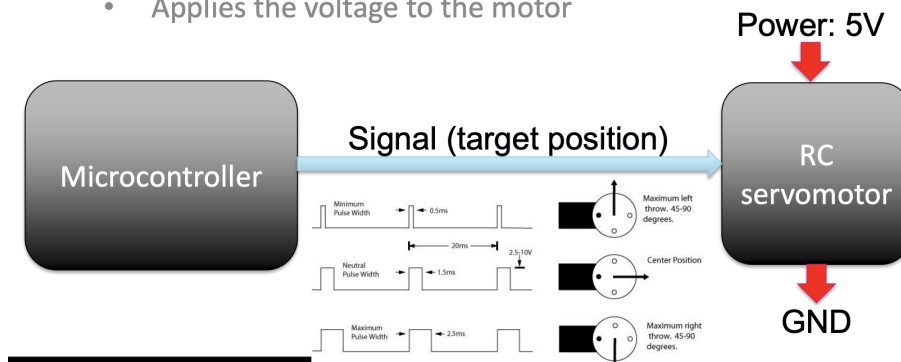


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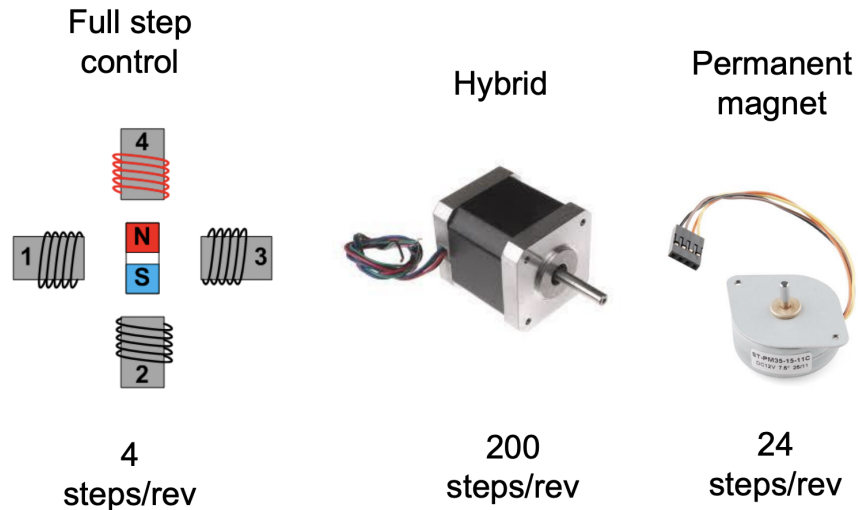
RC servos



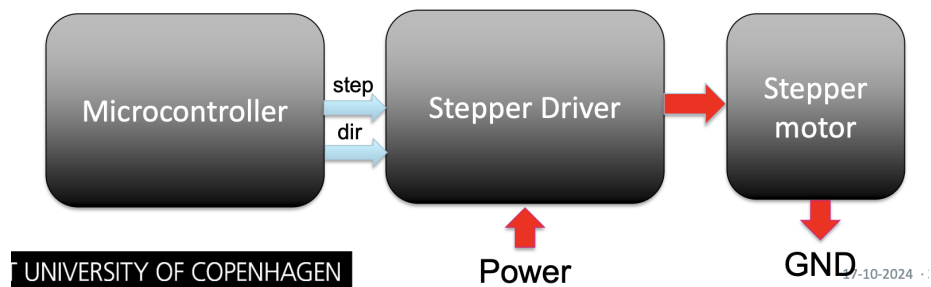
- Microcontroller sets a target
- Servomotor tries to reach the target
 - Calculates the error
 - Calculates the output (voltage)
 - Applies the voltage to the motor



Stepper Motor

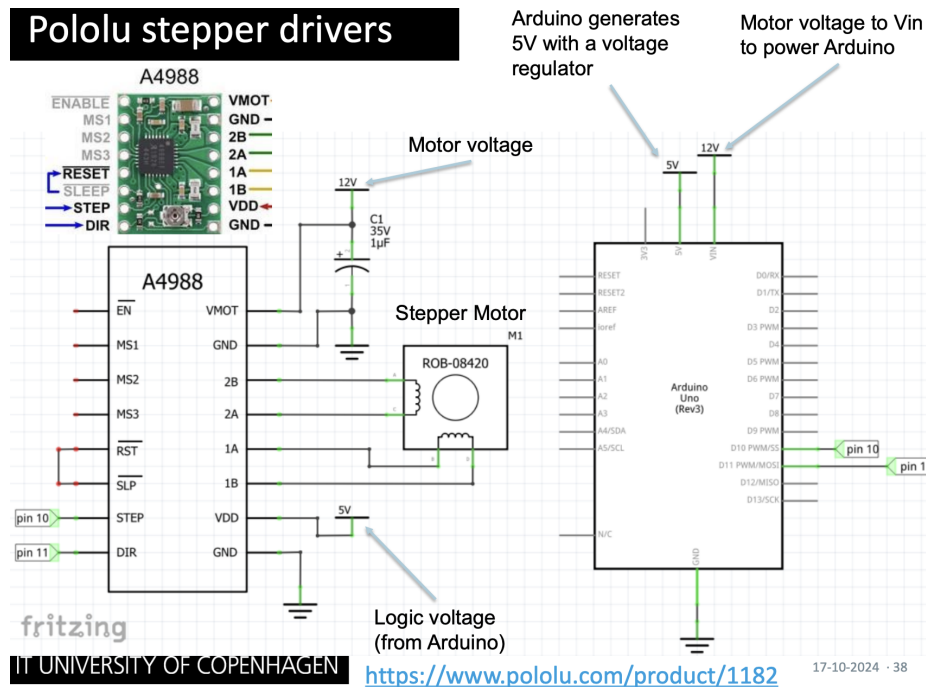


- Constant current for each step
- A step sequence generates the rotation
- Open loop control (if the force of the motor is enough!!)
- Microcontroller gives direction and step (pulses) signals
- Stepper driver handles the current through the coils



Important with a large (47 not 1) electrolytic capacitor for below circuit!

Pololu stepper drivers



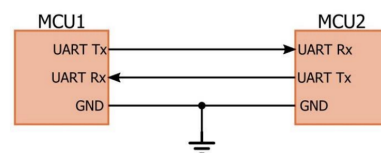
Communication Protocols

Sometimes we may want to communicate between multiple uControllers. This is done through *digital protocols*. Multiple devices are connected together in a *master-slave* type connection, where one device is the primary communicator and controller. It is also easy to implement and has medium latency (400 kHz - 5 MHz). There are multiple types, some of which are:

- **USART** - Universal Synchronous Asynchronous Receiver Transceiver (Serial) - 2 lines:

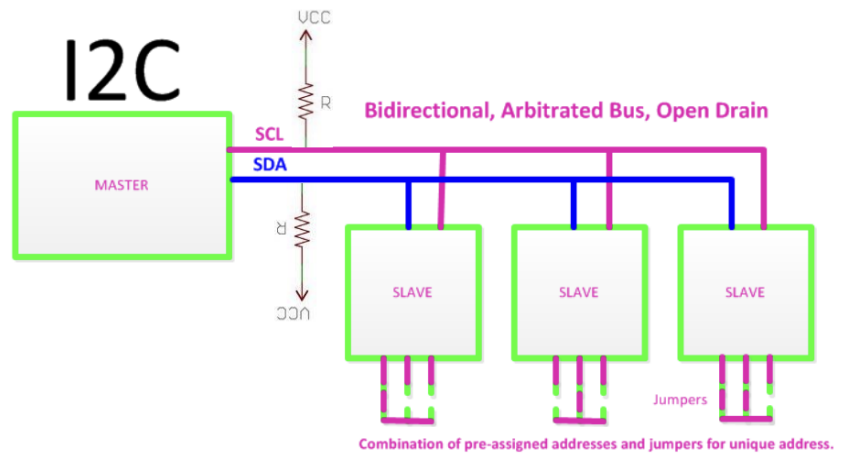
Rx: Reception (input)
Tx: Transmission (output)

TX are RX are crossed over!

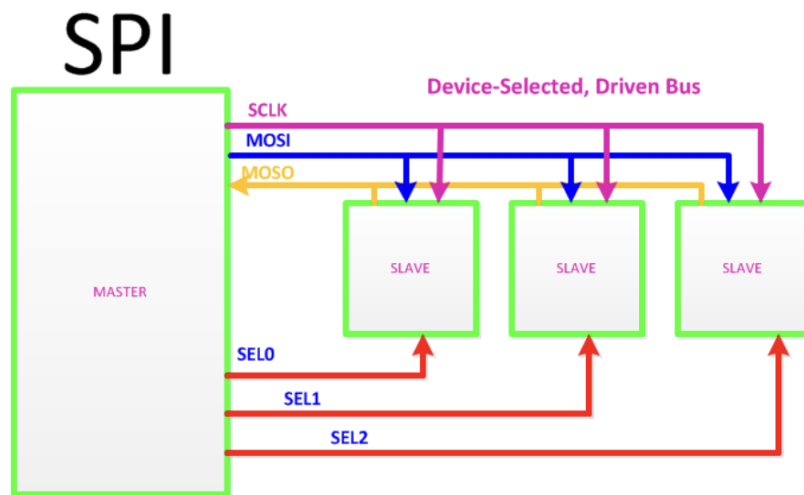


rial)

- F.ex. Computer to Arduino communication. (Only 2 devices in communication)
- Uses two lines on each communication device, *Rx* (Receiver) & *Tx* (Transmitter) that are crossed over to send signals.

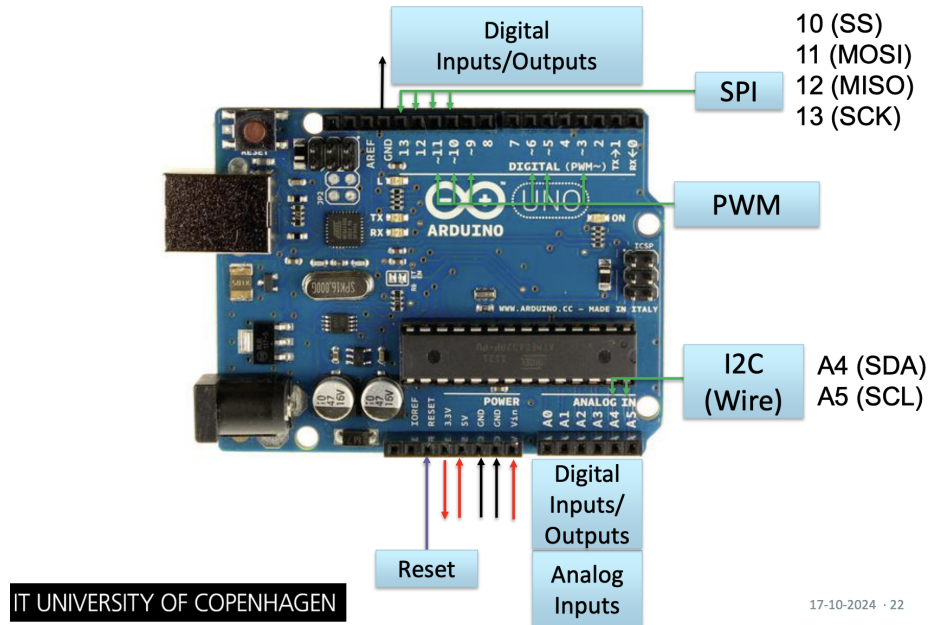


- **I2C (Wire or TWI)**
 - Uses two lines (SDA & SCL) with pull-up resistors.
 - Slaves have unique addresses and multiple can be connected to the same bus.



- **SPI**
 - uses three shared lines (MISO, MOSI, SCLK)
 - Slaves each use their own extra line and therefore does not need an address.

Arduino - Headers



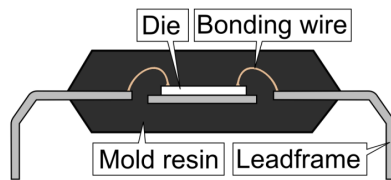
Microcontrollers: reading sensors, driving actuators, basic SPI/I2C

Microcontroller - IC (Integrated Circuit)

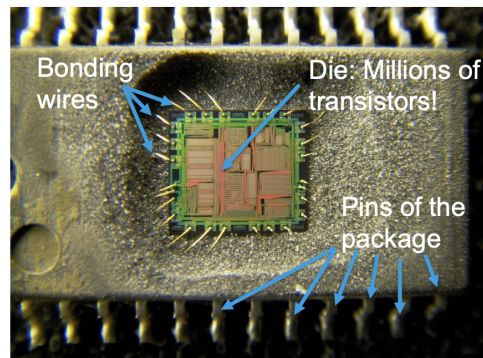
It contains all of the following components and function as a “mini”-computer / dedicated computer running a single program. - **CPU:** Main processor handling logic of the computer - **Memory:** RAM (short term variable values) & ROM (flash storage for program and other permanent storage) - **Inputs & Outputs:** Digital Pins (In & Out), Analog Pins (In), Timers, Serial Ports, Parallel Ports - **Internal Oscillator**

The IC works by connecting all of these components together through many transistors and provides relevant pins in an outwards interface that may be easily connected to PCBs and BreadBoards. See below image:

The physical package provides a way to easily connect the die of the microcontroller to a breadboard or Printed Circuit Board (PCBs)



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Public domain, via Wikimedia Commons

The ATmega328P is an example of an IC controller, and has the following specs and pin overview:

Microcontrollers – ATmega328P

One Integrated Circuit (IC), contains:

Processor (CPU)

Memory

ROM:

Program (Flash) 32KB
Permanent data (EEPROM) 1KB

RAM:

variables 2KB

Input/Output Peripherals

Digital Input / Output 23GPIO

Analog Inputs (A/D Converter) 8 channels

Timers 2 – 8 bit, 1 – 16 bit

Serial ports

UART 1

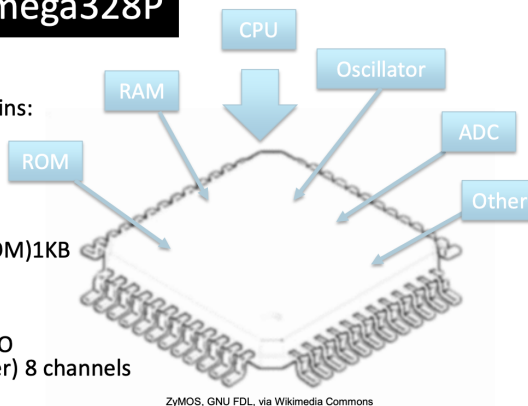
SPI 1

I2C 1

USB 0

Parallel ports

Internal Oscillator

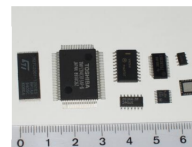


ZYMOS, GNU FDL, via Wikimedia Commons

Different physical packages available:



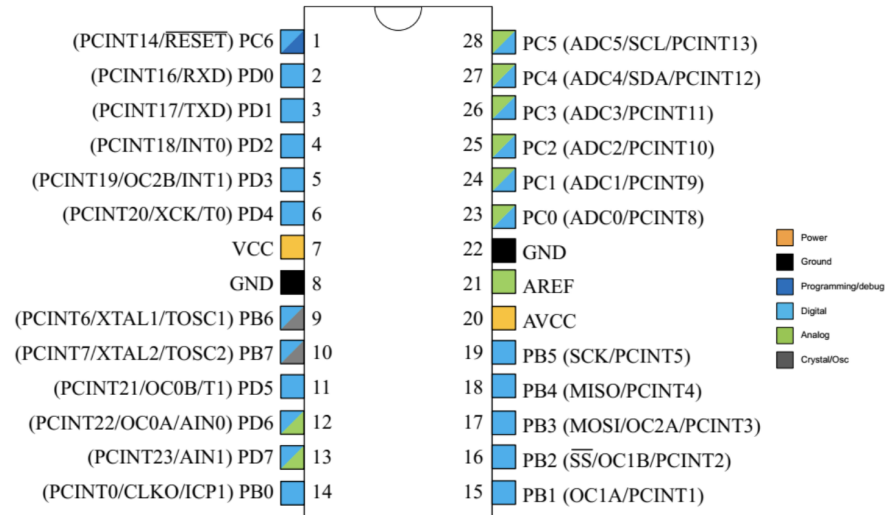
oomlout, CC BY-SA 2.0, via Wikimedia Commons



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Pin-out

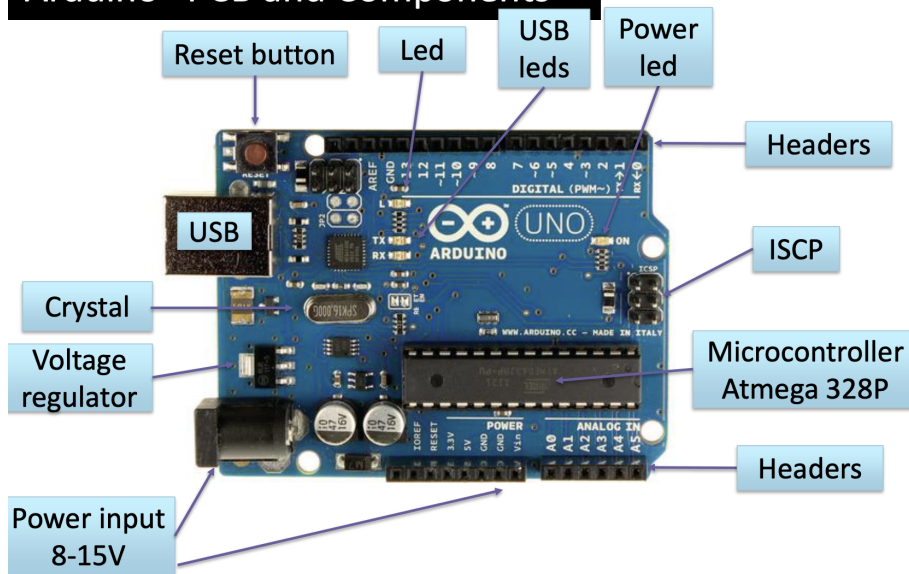
Figure 5-1. 28-pin PDIP



Arduino - uController

The Arduino is kind of like a “wrapper” around a micro controller, making it easier to interact with and control. The arduino is a *PCB with a microcontroller and connectors* - in particular it also adds an easier interface by providing a simple API and coding experience (USB porting and C++ coding).

Arduino - PCB and Components



Lecture statement: An Arduino is an electronic board with a microcontroller and expansion boards ready to use! (extension boards / PCBs are called *shields*)

Remember! to use transistors when driving motors with an Arduino, as the current is too high for the arduino to handle.

Inputs - From voltage to registers

When a voltage is applied by an **external circuit** to an Arduino's pin, it is **an input to the Arduino**. The pin must be set to an input. The voltage is translated to a register.

Digital – digitalRead(pin)

- 0V = 0 (LOW)
- 5V = 1 (HIGH)
- And 1V, 2.5V or 4V??????
From datasheet
 $1V = 0$
 $2.5V = 0/1$
 $4V = 1$

Analog – analogRead(pin)

- 0V = 0
- 5V = ?
- From datasheet
10bits ADC
 $5V = 2^{10}-1=1023$
- And 1V, 2.5V or 4V??????
 $1V = (1023/5) = 205$
 $2,5V (1023/2) = 511$
 $4V = (1023*4/5) = 818$

Inputs and Outputs

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Outputs - From registers to voltages

When the Arduino produces a voltage in a pin, it is an **Arduino's output**. The pin must be set to an output. The value of the register is translated to a voltage.

Digital – digitalWrite(pin, value)

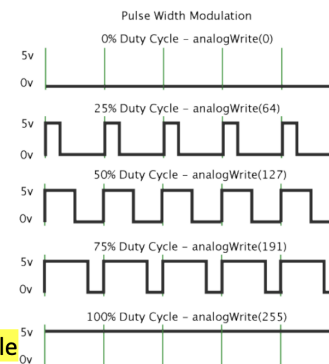
- 0 = 0V?? (**False!**)

From datasheet

- 0 = <1V
- 1 = >4.1V

Analog – analogWrite(pin, value)

- There is not a DAC in ucontroller
- Solution: Use PWM (Pulse Width Modulation) hardware, available at some pins
- Produce a digital signal oscillating from HIGH to LOW at a fixed frequency and control the duty cycle



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<https://www.arduino.cc/en/Tutorial/PWM>

Powering an Arduino

Connect an USB cable

The computer provides 5V to the 5V pin

Usually, cannot provide more than 0.5A (500mA)

Connect a 12V (or 9V) barrel plug

It provides 12V at the Vin pin

The Arduino voltage regulator transforms 12V to 5V

The voltage regulator can provide up to 0.5A

Connect a 5V battery or power supply to the 5V pin

Risk of damaging the components or your computer!

Be sure not to connect more than 5V!

4 alkaline batteries produce 6V when fully charged

Do not connect the USB cable at the same time

Power

PCBs: pros/cons vs breadboard, pads/tracks/vias, soldering basics

todo

Example Questions

Q1

todo

Q2

todo

Q3

Circuits to know by heart

1. Control a LED intensity with a potentiometer
2. Read a potentiometer (what is the position of the knob?), use a multimeter to read the voltage
3. Use a 3-pin switch and two LEDs. One LED should be always on, and changing the switch status changes the LED that is on

The Drink Dispenser